

Improving Preparedness of Communities for Evacuations using ZEVs

Workshop #2 With Community Partners

Modeling of a real-world evacuation in Mariposa, CA

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Outline

❑ Introduction

❑ A Microsimulation Platform in Aimsun for EV Charging and Evacuation

- Module for EVs and Fixed/Mobile Charging Stations
- Module for Vehicle Evacuation
- Module for Performance Evaluation

❑ Modeling of a real-world evacuation in Mariposa, CA

- The Mariposa Network
- Existing and Potential Charging Stations
- Three Case Studies with Synthetic Demand

❑ Next Steps

Part I: Introduction

Current Challenges

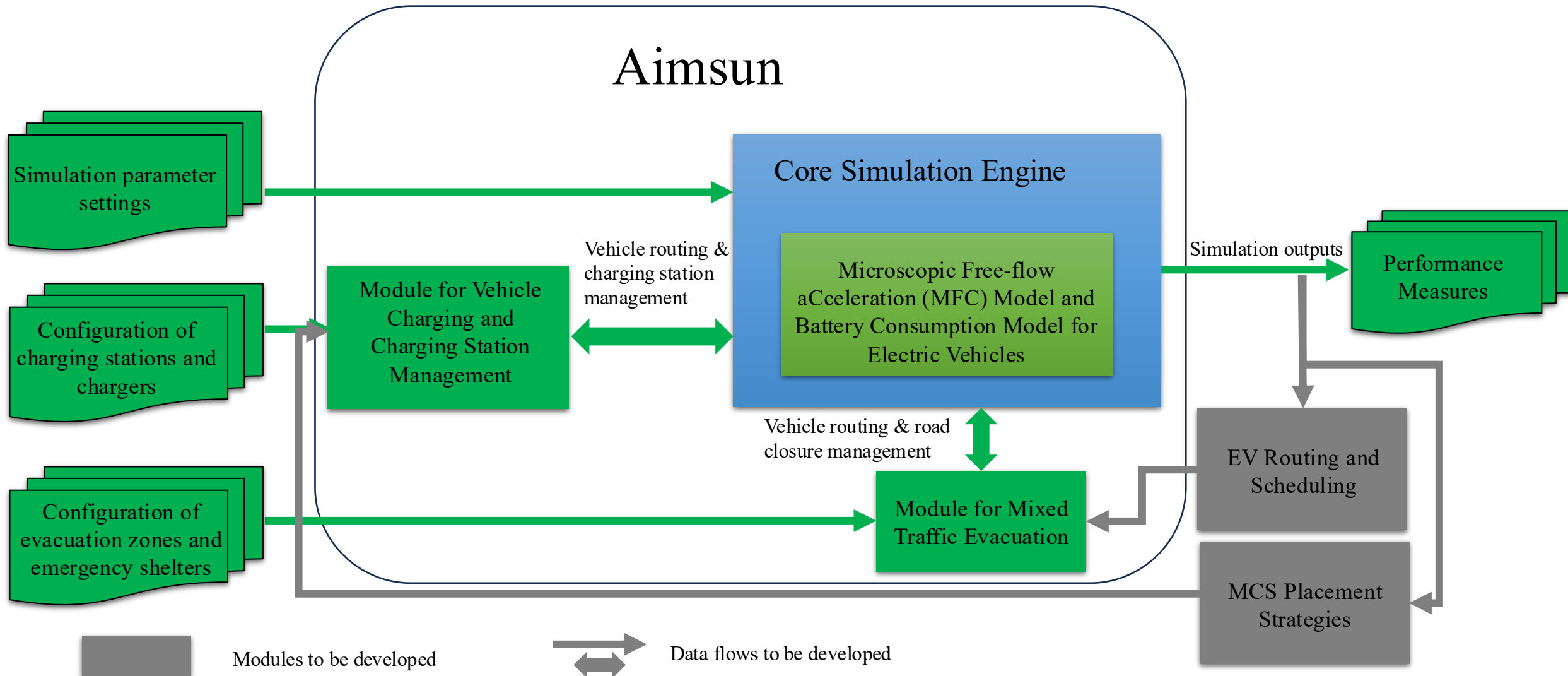
- ❑ Regulated by CA Executive Order N-79-2, we expect 100% sales of light-duty Zero Emission Vehicles (ZEV) by 2035.
- ❑ However, there are a lot of challenges that need to be addressed to support the statewide transition to ZEVs.
 - Is the power grid able to support the charging of a large amount of EVs, especially when natural hazards occur (e.g., wildfire, flood, and extreme heat)?
 - How many more charging stations are needed? Where should we place these new charging stations?
 - Are we ready to deal with evacuations using EVs? What could be the challenges?
 - How to minimize charging time/queuing during the evacuation of EVs? Mobile charging stations as an option?
 - How can we evaluate the performance of the proposed charging station placement strategies and evacuation plans for EVs?
 - Etc.

Our Research Goals

- ❑ State-of-the-art studies on the placement of charging stations and evacuation plans usually rely on the macroscopic-level planning and/or optimization tools.
 - There is a lack of a performance evaluation platform at a high granularity with more practical settings in penetration rates of EVs, charger powers, availability of chargers, battery capacities/ranges, temperatures, road configurations, and traffic conditions.
- ❑ In this research, we want to
 - Develop a microsimulation platform for EV charging and evacuation that can support various needs for stakeholders.
 - More realistic assessment of network/link/TAZ-level energy needs for EVs, e.g., Charging Corridor vs. Charging Communities
 - Identification of hotspots/bottlenecks of EV charging demand & performance assessment of the placement of EV charging stations
 - Performance assessment of EV evacuation strategies/plans, especially with Mobile Charging Stations (MCSs)
 - Develop a real-world evacuation scenario, e.g., in Mariposa, CA, to demonstrate the value of the developed microsimulation evaluation platform.

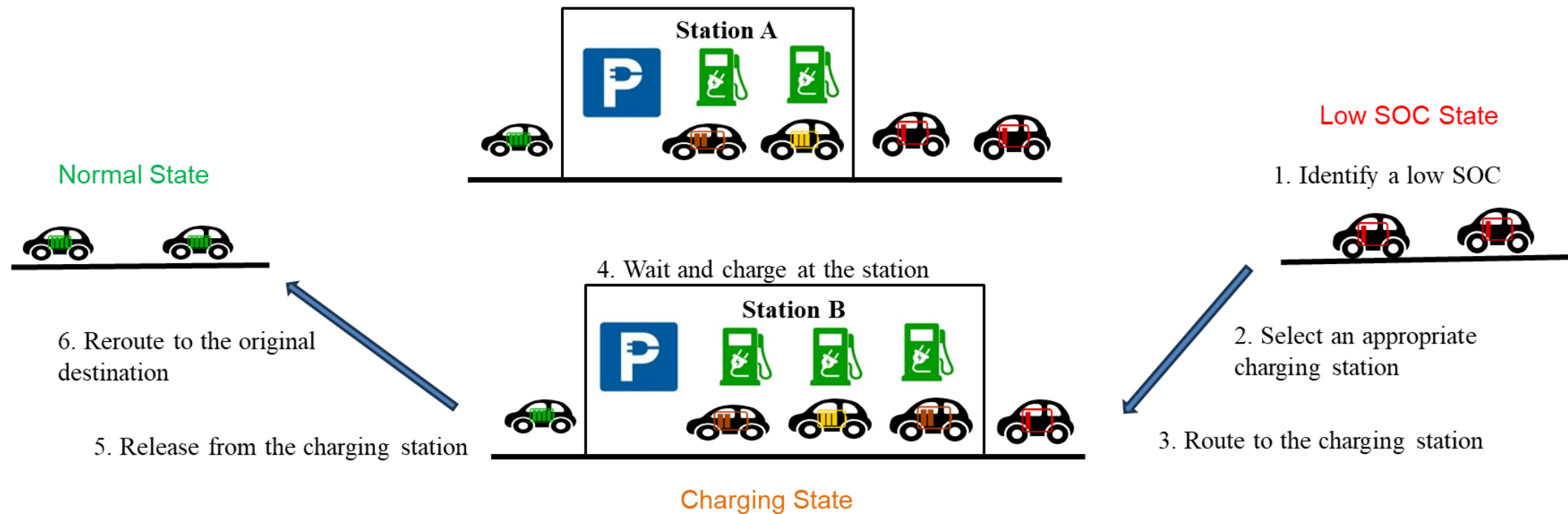
Part I: A Microsimulation Platform for EV Charging and Evacuation

System Design



Module for EV Charging and Charging Station Management (1/2)

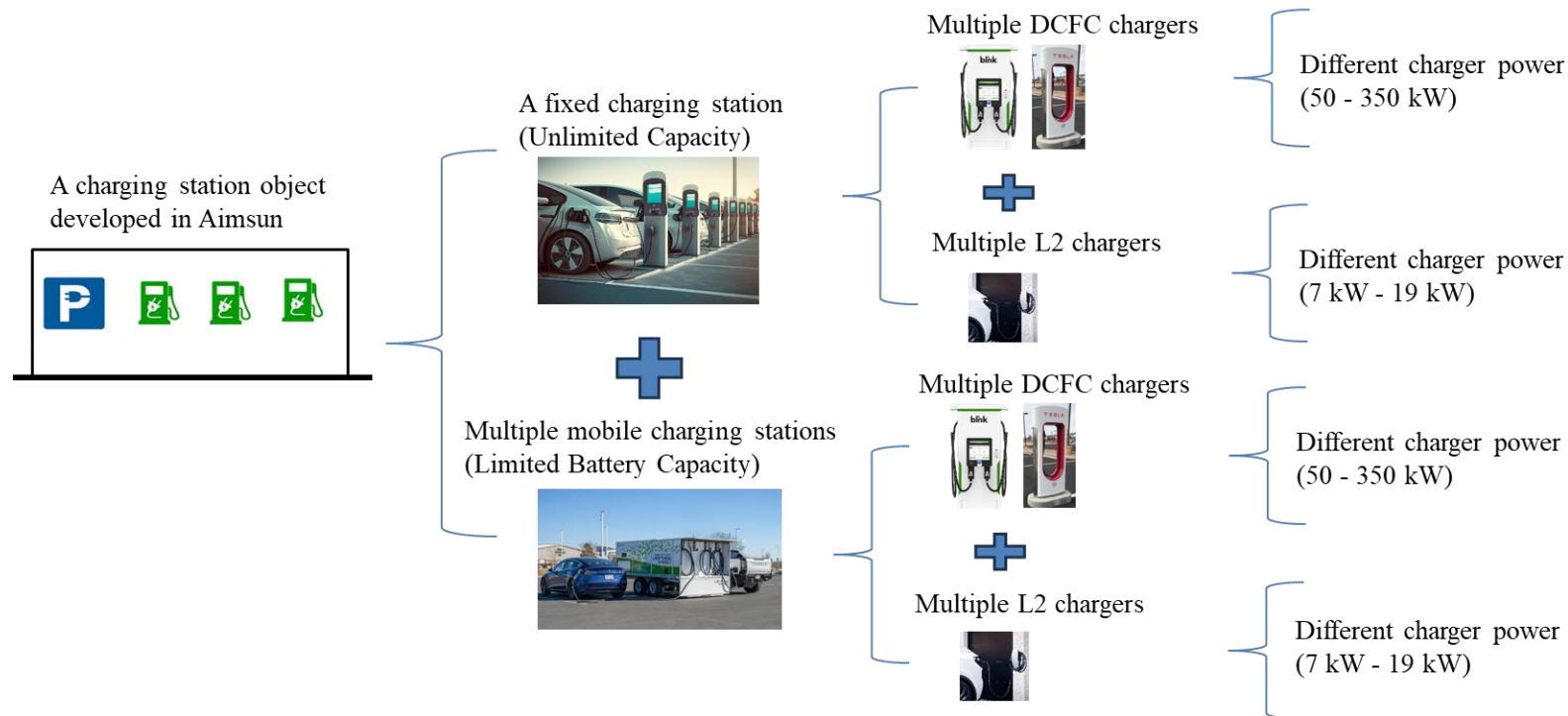
□ This module can replicate the real-world EV charging process.



Module for EV Charging and Charging Station Management (2/2)

□ This module is flexible to configure the charging stations with the following:

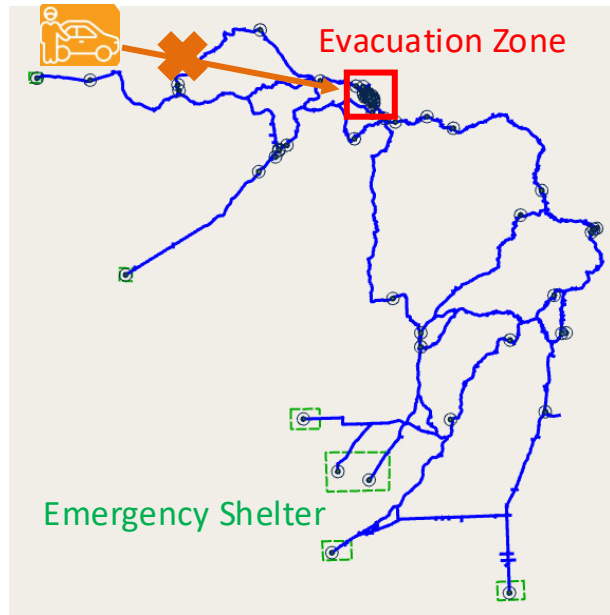
- Fixed and/or mobile charging stations → Modeled as centroids in Aimsun
- Different charger types
- Limited space with a maximum number of EVs allowed



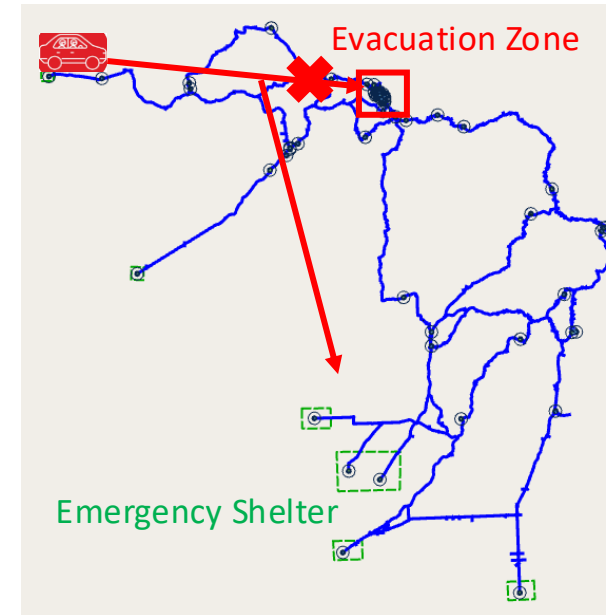
Module for Mixed Traffic Evacuation (1/2)

- ❑ This module supports the simulation of vehicle evacuations in a mixed traffic environment.

Case 1: vehicles that have destinations in the evacuation zone(s) but are not yet assigned will not enter the network.

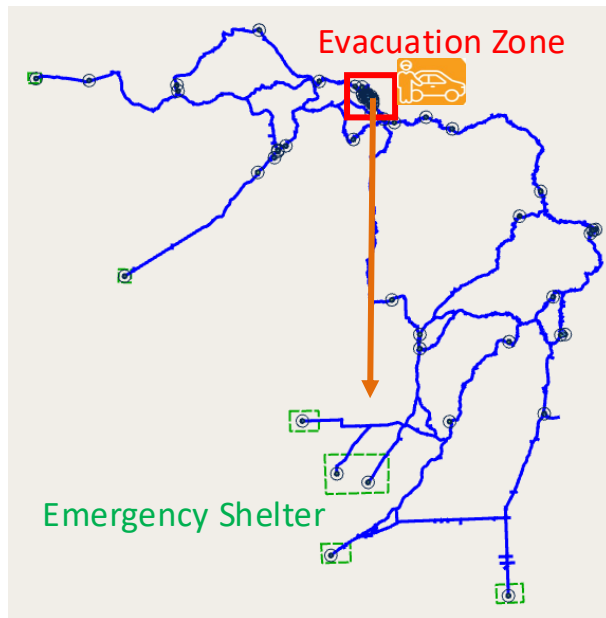


Case 2: vehicles that have destinations in the evacuation zone(s) and are already in the network will be detoured to one of the nearest emergency shelters.



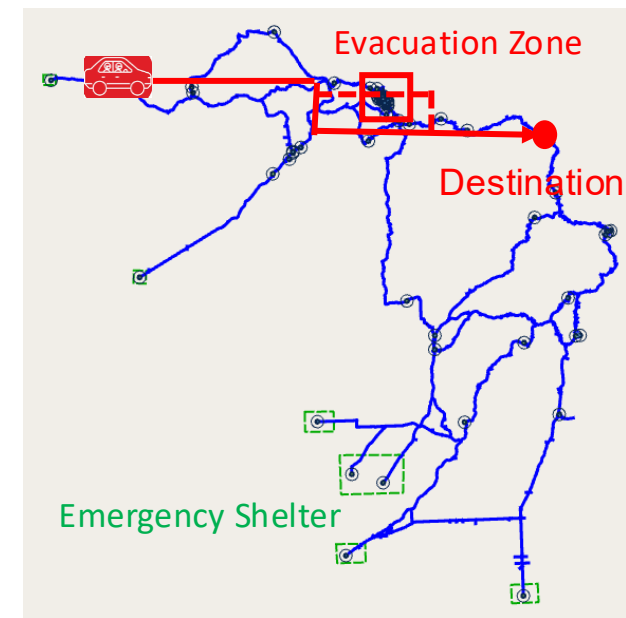
Module for Mixed Traffic Evacuation (2/2)

Case 3: vehicles that have origins in the evacuation zone(s) will be detoured to one of the nearest emergency shelters.



This module can enable a certain percentage of vehicles to follow the evacuation routes if available.

Case 4: vehicles that have paths crossing in the evacuation zone(s) will be detoured to bypass the evacuation zone(s).



Module for Performance Measures

□ This module outputs the following measures and statistics to assess the performance of charging station placement strategies and evacuation plans.

- **Real-time charging station status**

- DCFC/L2 EVs served/being charged/waiting, DCFC/L2 chargers available/being used/inactive, current battery consumptions for fixed/mobile charging stations, etc.

- **Real-time link status**

- Time, average speed/flow/density in the last time interval

- **Statistics**

- **Network performance summary:** *Number of vehicles/EVs entered, network battery consumption, number of EVs failed to find a charging station/run out of battery, total network VMT/VHT, etc.*
- **Charging station summary:** *DCFC/L2 EVs served/being charged/waiting, total battery consumptions for fixed/mobile charging stations, etc.*
- **Link battery consumption summary:** *Average speed/flow/density, total travel distance/time, total battery consumed, etc.*
- **EV exit statistics:** *Initial/final SOC, battery consumed, total travel distance, etc.*
- Log information for EVs failed to find a charging station

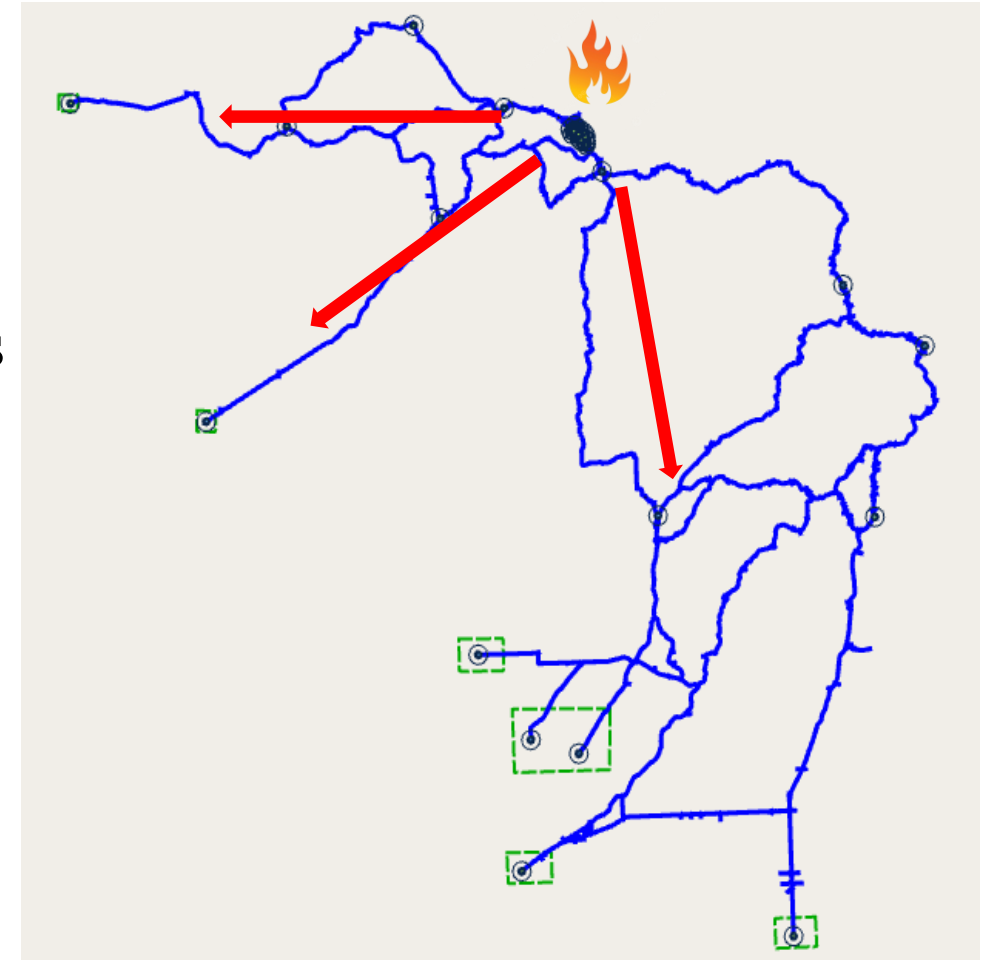
Part II: Modeling of a real-world evacuation in Mariposa, CA

Why Evacuation Use Cases for Mariposa, CA?

- ❑ Due to the rapid increase in ZEV ownership in recent years, rural areas like in Mariposa are facing a big challenge in the evacuation using ZEVs.
 - These rural areas normally lack charging facilities to support mass evacuations using ZEVs.
 - The transportation networks are vulnerable to mass evacuations.
 - Very limited routes to nearby cities.
 - These routes normally consist of state highways that are 2-lane roadways. Therefore, any traffic incident happening on the roadway will significantly degrade the evacuation effectiveness and efficiency.
 - Changes to traffic control at arterial intersections are needed to support the large volume of vehicles traveling in the dominated direction.

The Mariposa Network

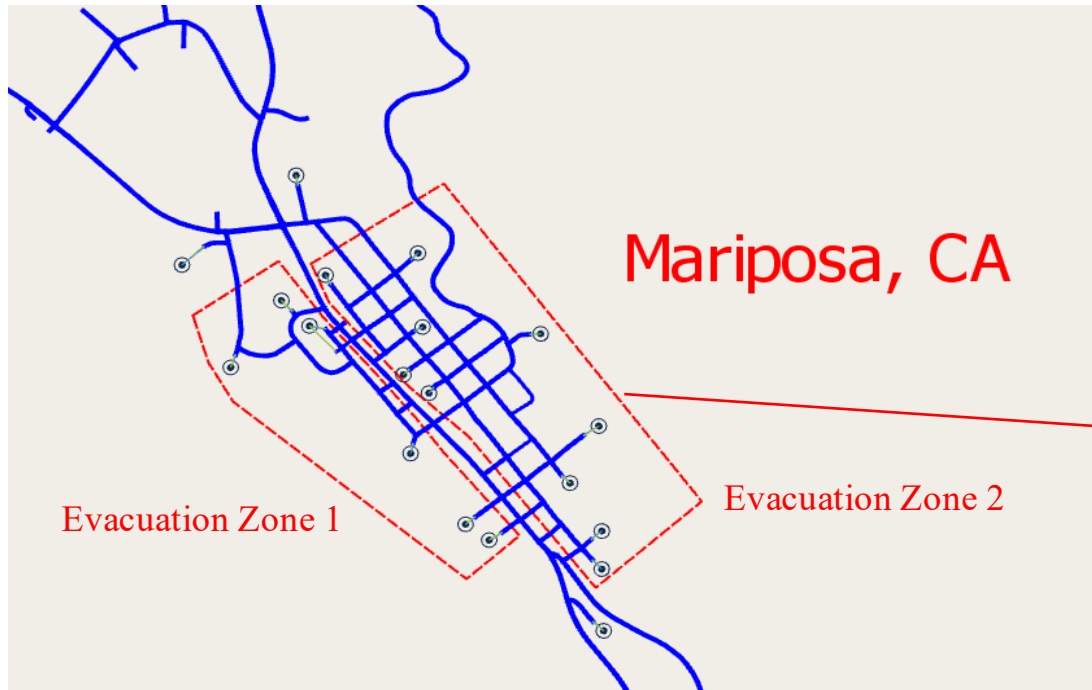
- ❑ We modeled the major arterials in the town of Mariposa, the state highways, and other 2-lane roadways.
 - ❑ Road geometry and speed limits have been updated to match those in the field.
- ❑ We consider the scenario of evacuating the residents in the town of Mariposa, CA to a safer place (to nearby cities in the central valley), due to a natural disaster (e.g., wildfire).
- ❑ Key components
 - ❑ Evacuation Zones
 - ❑ Emergency Shelters
 - ❑ Charging Stations
 - ❑ *Normal & Evacuation OD Demand, Signal Control Settings (On-Going)*



Setting of Evacuation Zones

□ The developed platform supports the modeling of multiple evacuation zones.

- For example, we created two evacuation zones in the town of Mariposa, CA.



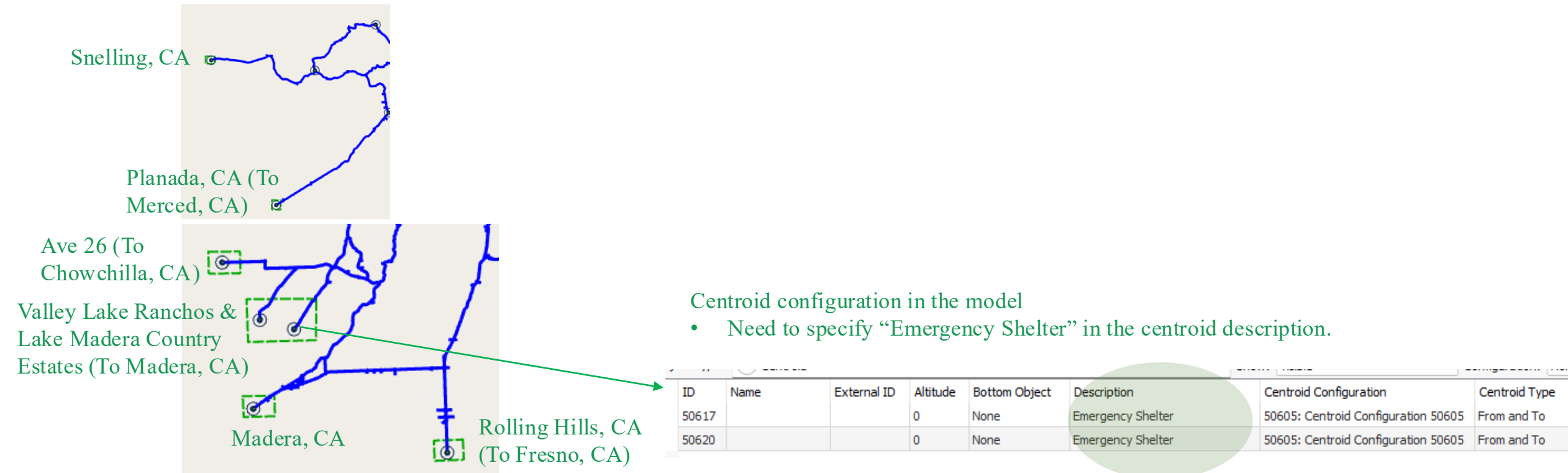
Centroid configuration in the model

- Need to specify “Evacuation Zone” in the centroid description.

ID	Name	External ID	Altitude	Bottom Object	Description	Centroid Configuration	Centroid Type
50653			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50677			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50656			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50643			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50674			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50671			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To
50680			0	None	Evacuation Zone	50605: Centroid Configuration 50605	From and To

Setting of Emergency Shelters

- ❑ The developed platform also supports the modeling of multiple emergency shelters.
 - For example, we created six emergency shelters (destinations) in the study area.



Setting of Charging Stations (1/4)

❑ Currently there are only 12 public charging stations within the study network.

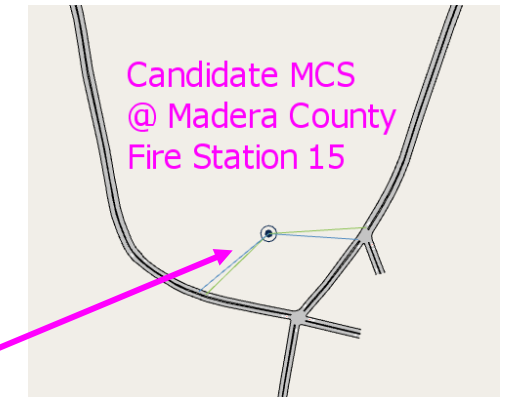
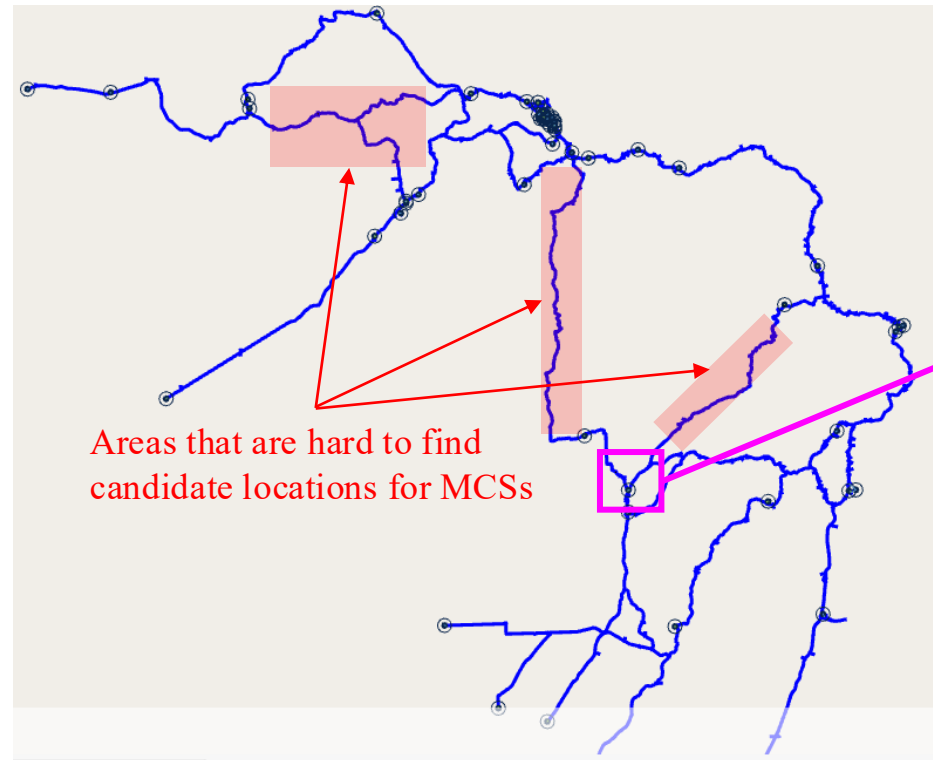
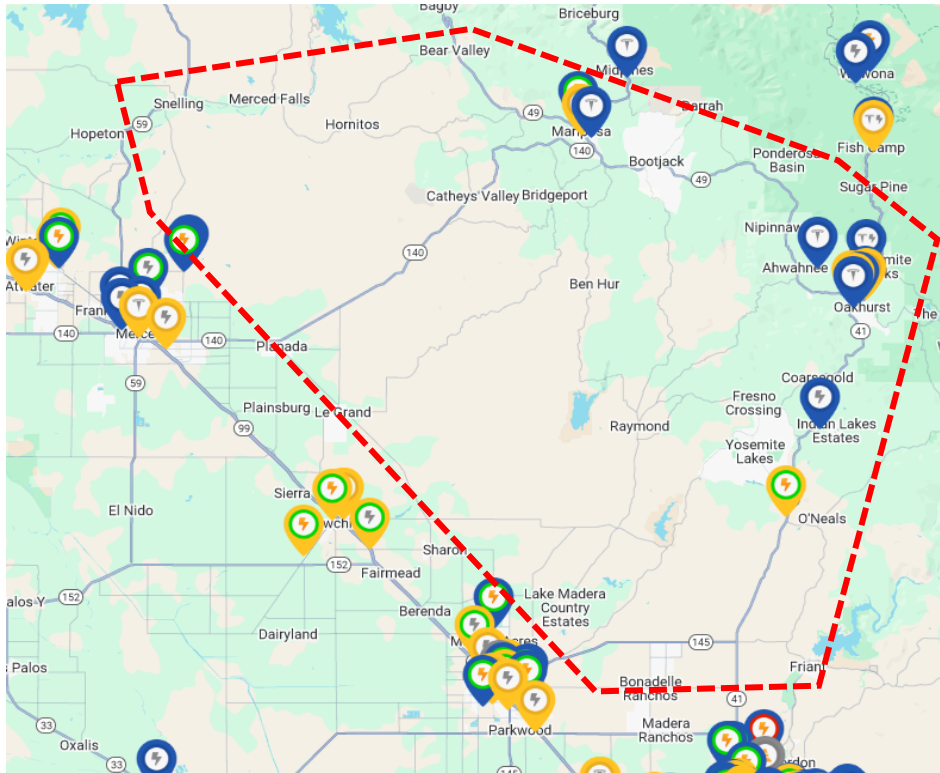
- A total of 42 L2 chargers
- A total of 34 DC Fast chargers

Station #	Name	Location	City	Location Type	Type	Plugs	Ports	Charger Power (KW)	GeoLocation
1	MACT MARIPOSA	5204 Hospital Rd, Mariposa, CA 95338	Mariposa, CA	Hospital	L2	EV Plug/J1772	2	6.5	37.50158041074161, -119.97682852883486
2	Pioneer Market - Tesla Supercharger	5034 Coakley Circle Mariposa, CA 95338	Mariposa, CA	Grocery Store	DC Fast	Tesla/J3400	12	250	37.48861764643503, -119.97188094566519
3	Yosemite Plaisance Bed & Breakfast - Tesla Destination	4953 Trabucco St Mariposa, CA 95338	Mariposa, CA	Hotel	L2	Tesla/J3400	2	16	37.48663710774416, -119.96136849325065
4	Oakhurst Library Parking Lot	49044 Civic Cir Oakhurst, CA 93644	Oakhurst, CA	Library	L2	EV Plug/J1772	4	6.5	37.33104691165413, -119.65333536100233
5	Best Western Plus - Yosemite Gateway Inn	40530 Highway 41 Oakhurst, CA 93644	Oakhurst, CA	Hotel	L2	EV Plug/J1772	2	6.5	37.33553291220319, -119.64639858984503
					L2	Tesla/J3400	2	13	37.33553291220319, -119.64639858984503
6	Oakhurst, CA Supercharger	40489 CA-41, Oakhurst, CA, 93644	Oakhurst, CA	Hotel	DC Fast	Tesla/J3400	8	250	37.33462063082841, -119.64564738984507
7	Chargepoint Charging Station, Old Mill Village	40050 CA-49, Oakhurst, CA, 93644	Oakhurst, CA	Plaza	DC Fast	CHAdEMO	2	62.5	37.33202347007937, -119.65824890063992
					DC Fast	CCS	2	62.5	
					L2	EV Plug/J1772	2	6.5	
8	Electrify America Charging Station, Central Valley Bank	40004 CA-41, Oakhurst, CA, 93644	Oakhurst, CA	Bank	DC Fast	CHAdEMO	1	50	37.32994595637176, -119.65767955152968
					DC Fast	CCS	3	150	
					DC Fast	CCS	4	350	
9	Chateau du Sureau - Tesla Destination	48688 Victoria Lane Oakhurst, CA 93644	Oakhurst, CA	Hotel	L2	Tesla/J3400	2	16	37.32740278041003, -119.66156961717093
10	Chuckchansi Resort & Casino	711 Lucky Ln Coarsegold, CA 93614	Coarsegold, CA	Hotel	L2	EV Plug/J1772	4	7	37.211498616649855, -119.69916997836019
					L2	Tesla/J3400	9	10	
					L2	EV Plug/J1772	9	10	
11	Chukchansi Crossing Fuel Station & Travel Center	32025 CA-41 Coarsegold, CA 93614	Coarsegold, CA	Fuel Station	L2	Tesla/J3400	2	16	37.214480562925154, -119.70790409264072
					L2	EV Plug/J1772	2	7	
12	Blink Charging Station, Chawanakee Unified School District	26065 Outback Industrial Way, O'Neals, CA, 93645	O'Neals, CA	School	DC Fast	CCS	2	120	37.12903309555086, -119.7334498895609

References: <https://chargehub.com/map/#/en/map>; <https://www.tesla.com/findus>; <https://afdc.energy.gov/fuels/electricity-locations#/find/nearest?fuel=ELEC>

Setting of Charging Stations (2/4)

- ❑ A large area of the study network is not covered by any charging station.
 - We added candidate locations (e.g., fire station, fuel station, hotel, grocery store/plaza, and school/library) that have enough parking space to support the deployment of Mobile Charging Stations (MCSs).



Setting of Charging Stations (3/4)

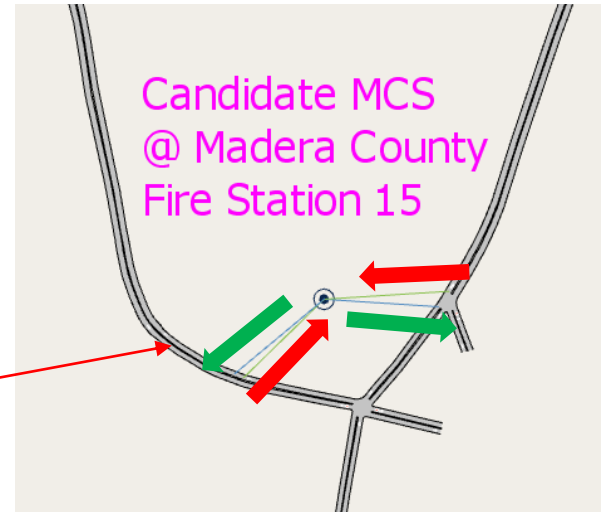
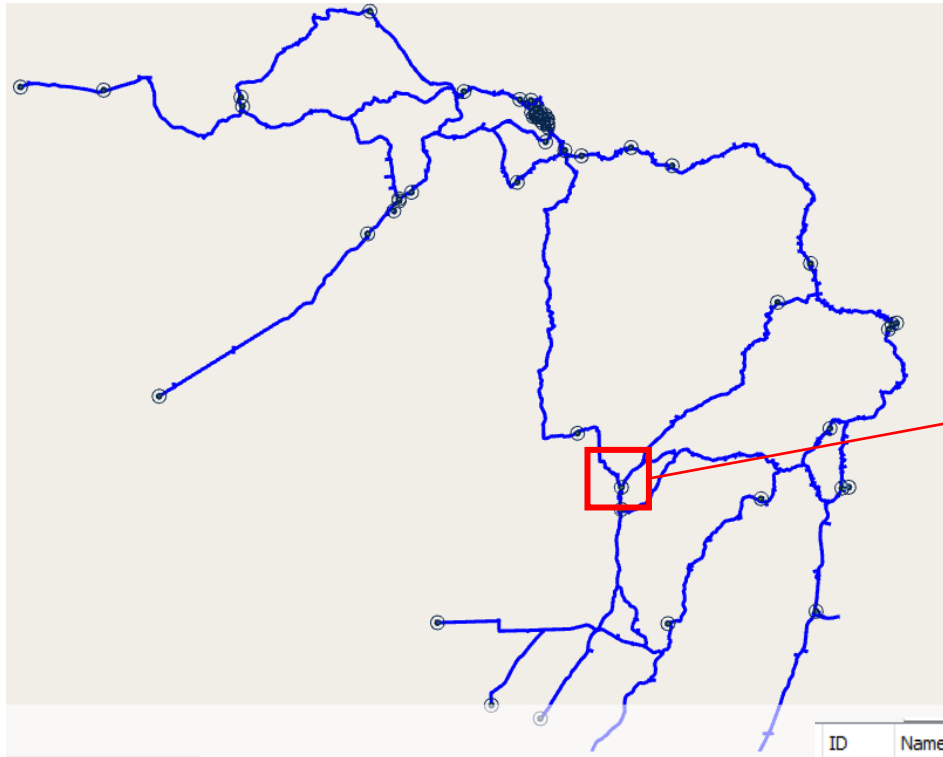
❑ A total of 34 centroids were created for existing and candidate charging stations.

- 9 centroids for existing charging stations
- 25 centroids for candidate mobile charging stations

ID	Name	Description	Centroid Configuration	Centroid Type
51107	Yosemite Plaisance Bed & Breakfast - Tesla Destination	Charging Station	50605: Centroid Configuration 50605	From and To
51094	Pioneer Market - Tesla Supercharger	Charging Station	50605: Centroid Configuration 50605	From and To
51078	MACT MARIPOSA Charging Station	Charging Station	50605: Centroid Configuration 50605	From and To
51172	Charging Station @ Chuckchansi Resort & Casino	Charging Station	50605: Centroid Configuration 50605	From and To
50787	Chargepoint & Electrify America Charging Stations	Charging Station	50605: Centroid Configuration 50605	From and To
51398	Candidate MCS @Valero	Charging Station	50605: Centroid Configuration 50605	From and To
51408	Candidate MCS @Sierra Foothill Charter School	Charging Station	50605: Centroid Configuration 50605	From and To
50695	Candidate MCS @Oasis	Charging Station	50605: Centroid Configuration 50605	From and To
51372	Candidate MCS @Mariposa County Fire Department	Charging Station	50605: Centroid Configuration 50605	From and To
50725	Candidate MCS @Mariposa Co. Fire Station 25 Mt. Bullion/Airport	Charging Station	50605: Centroid Configuration 50605	From and To
51434	Candidate MCS @Forestry & Fire Protection	Charging Station	50605: Centroid Configuration 50605	From and To
50734	Candidate MCS @Church of St. Catherine of Siena	Charging Station	50605: Centroid Configuration 50605	From and To
51388	Candidate MCS @Cathey's Valley Park	Charging Station	50605: Centroid Configuration 50605	From and To
50753	Candidate MCS @California Historical Landmark No. 331	Charging Station	50605: Centroid Configuration 50605	From and To
51422	Candidate MCS @CAL FIRE MMU Catheys Valley Fire Station	Charging Station	50605: Centroid Configuration 50605	From and To
51460	Candidate MCS @CAL FIRE MMU Hornitos Fire Station	Charging Station	50605: Centroid Configuration 50605	From and To
51504	Candidate MCS @Anglers Edge Market	Charging Station	50605: Centroid Configuration 50605	From and To
51362	Candidate MCS @ Mariposa County Fire Dept. Station 28	Charging Station	50605: Centroid Configuration 50605	From and To
51217	Candidate MCS @ Lutheran Church - Mariposa	Charging Station	50605: Centroid Configuration 50605	From and To
50778	Candidate MCS @ Wasuma Elementary School & Fresno Flats Comm Day School	Charging Station	50605: Centroid Configuration 50605	From and To
51237	Candidate MCS @ Valero Gas Station	Charging Station	50605: Centroid Configuration 50605	From and To
51315	Candidate MCS @ Rivergold Elementary School	Charging Station	50605: Centroid Configuration 50605	From and To
51351	Candidate MCS @ Raymond Bridge Trailhead For Lakeview Trail	Charging Station	50605: Centroid Configuration 50605	From and To
50797	Candidate MCS @ Madera County Fire Station 15	Charging Station	50605: Centroid Configuration 50605	From and To
51227	Candidate MCS @ Kwik Serv	Charging Station	50605: Centroid Configuration 50605	From and To
51248	Candidate MCS @ Homestead Cottages	Charging Station	50605: Centroid Configuration 50605	From and To
51305	Candidate MCS @ Coarsegold Elementary School	Charging Station	50605: Centroid Configuration 50605	From and To
51276	Candidate MCS @ CAL FIRE MMU Raymond Fire Station	Charging Station	50605: Centroid Configuration 50605	From and To
51325	Candidate MCS @ Buck Ridge	Charging Station	50605: Centroid Configuration 50605	From and To
51193	Candidate MCS @ California State Mining & Mineral Museum	Charging Station	50605: Centroid Configuration 50605	From and To
51145	Chateau du Sureau - Tesla Destination	Charging Station	50605: Centroid Configuration 50605	From and To
50807	Chukchansi Crossing Fuel Station & Travel Center	Charging Station	50605: Centroid Configuration 50605	From and To
51135	Charging Stations@Oakhurst Library@Best Western Plus@Oakhurst Supercharger	Charging Station	50605: Centroid Configuration 50605	From and To
51183	Blink Charging Station @ Chawanakee Unified School District	Charging Station	50605: Centroid Configuration 50605	From and To

Setting of Charging Stations (4/4)

❑ We modeled charging stations as centroids



Centroid configuration in the model

- Need to specify “Charging Station” in the centroid description.

ID	Name	External ID	Altitude	Description	Centroid Configuration	Centroid Type
50797	Candidate MCS @ Madera County Fire Station 15		0	Charging Station	50605: Centroid Configuration 50605	From and To

A Test Scenario with Synthetic Demand

❑ Three vehicle types

- Car
- Car – EV DCFC → Can use both DCFC and L2 chargers
- Car – EV L2 → Can only use L2 chargers

❑ 1-hr simulation (08:00-09:00)

- 1-hr synthetic traffic demands (a total of 3000+ vehicles)
 - 0%EV, 10%EV, 20%EV, 30%EV, 40%EV, 50%EV
 - For EVs, 80% Car – EV DCFC and 20% Car – EV L2
 - 30-min warm-up period
- Initial SOC distribution
 - High at-home charge rate: Normal distribution with mean 80% and deviation 30%
 - Low at-home charge rate: Normal distribution with mean 50% and deviation 30%
- Charging Stations
 - Baseline scenario: with only fixed charging stations
 - Alternative scenario: with Mobile Charging Stations added
- SOC Thresholds
 - LOW SOC: 20%
 - Targeted SOC: 80%
- Evacuation (if enabled) starts at 8:15AM and ends at 9:00AM

Case Study: Induced VMT, Energy Demand, etc.

□ A baseline scenario

- With existing fixed charging stations
- High at-home charge rate ~ N(80%,30%)
- No evacuation activated

Results Averaged Over 5 Simulation Runs						
EV Percentage	0%	10%	20%	30%	40%	50%
Total VMT (miles)	45001	45229	46204	46118	46411	44323
Energy Demand (KWH)	0	784.33	1621.28	2504.84	3329.22	4057.56
# of EVs Failed to Find a Charging Station	0	13	27	44	52	62
# of EVs Run Out of Battery*	0	1	7	11	11	14

Increased VMT due to the need to look for a charging station

Increased energy needs due to the rapid increase in EV ownership

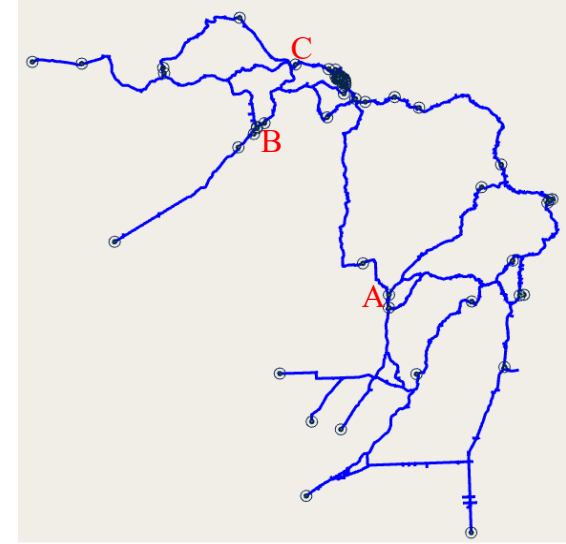
A higher failure rate as the percentage of EVs increases.

**EVs that run out of battery will be removed from the network/simulation.*

Case Study: How Many Charging Stations are Needed? Where to Place Them?

❑ An alternative scenario: With three fixed charging stations added

- Location A: @Madera County Fire Station 15 → 5 DCFC chargers at 250KW
- Location B: @Valera Gas Station → 5 DCFC chargers at 250KW
- Location C: @Mariposa Co. Fire Station 25 Mt Bullion/Airport → 10 DCFC chargers at 250KW



Results Averaged Over 5 Simulation Runs						
EV Percentage		10%	20%	30%	40%	50%
Total VMT (miles)	Baseline Scenario	45229	46204	46118	46411	44323
	Alternative Scenario	44944	45859	46601	46508	45933
Energy Demand (KWH)	Baseline Scenario	784.33	1621.28	2504.84	3329.22	4057.56
	Alternative Scenario	792.97	1611.15	2513.48	3312.24	4125.68
# of EVs Failed to Find a Charging Station	Baseline Scenario	13	27	44	52	62
	Alternative Scenario	6	16	19	29	34
# of EVs Run Out of Battery*	Baseline Scenario	1	7	11	11	14
	Alternative Scenario	2	6	8	10	13

The increase in charging station availability will reduce the need to travel a long distance for charging.

The increase in charging station availability will increase the chance for an EV to find a charging station nearby.

*EVs that run out of battery will be removed from the network/simulation.

Case Study: Evacuation with/without Mobile Charging Stations (1/2)

□ Proposed Evacuation Event

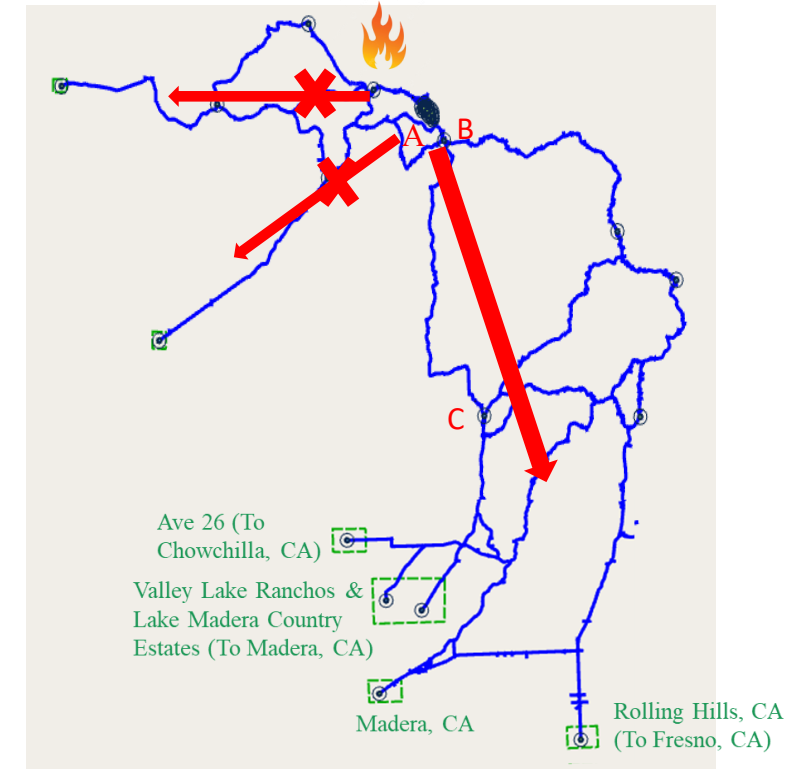
- A wildfire near Mariposa forces the residents to evacuate to the south (Madera, CA)

□ Global settings

- 20% EV
- Low at-home charge rate in the evacuation zone(s) ~ N(50%, 30%)
- High at-home charge rate for other locations ~ N(80%, 30%)
- Evacuation starts at 8:15AM, till the end of simulation (9AM)
- SOC Thresholds: Low SOC at 30%, Targeted SOC at 60%

□ Scenarios

- A baseline scenario: With existing fixed charging stations
- An alternative scenario: Three 600KWH mobile charging stations with 4 DCFC ports @ 150KW and 2 L2 ports @ 16KW
 - Location A @Mariposa County Fire Department
 - Location B @California State Mining & Mineral Museum
 - Location C @Madera County Fire Station 15



Case Study: Evacuation with/without Mobile Charging Stations (2/2)

20% EV & Results Averaged Over 5 Simulation Runs		
Total VMT (miles)	Baseline Scenario	37732.95
	Alternative Scenario	38230.3
Energy Demand (KWH)	Baseline Scenario	1436.3
	Alternative Scenario	1499.69
# of EVs Failed to Find a Charging Station	Baseline Scenario	63
	Alternative Scenario	22
# of EVs Run Out of Battery*	Baseline Scenario	7
	Alternative Scenario	4

The deployment of MCSs significantly increases the chance for an EV to be charged in a timely manner and thus enhances the efficiency and resilience of the transportation system during emergency evacuations.

*EVs that run out of battery will be removed from the network/simulation.

Mobile Charging Station Statistics (At the End of Simulation)					
Station Name	Battery Consumed (KWH)	# of EVs Served	# of EVs Being Charged	# of EVs Waiting	# of EVs In Route
@Mariposa County Fire Department	108	8	4	0	5
@California State Mining & Mineral Museum	32	3	2	0	7
@Madera County Fire Station 15	115	8	3	0	9

Part III: Next Steps

Next Steps: Model Calibration

□ Data Requirements

- PeMS Vehicle Detector Station (VDS) data
- Average Daily Traffic (ADT) data (probably from PeMS)
- OD demand data from regional travel demand models (Need help on this)
- Household data and car ownership data in the town of Mariposa (Need help on this)
- A few traffic signal control plans (less than 10 in our network; need help on this)

Next Steps: Strategies for MCS Placement

- We will implement strategies that dynamically allocate/place MCSs to the candidate locations to minimize the total/average evacuation time, based on the real-time information of
 - Network traffic status
 - Road closures
 - Status of charging stations
 - Etc.

Thanks!

Why Aimsun?

□ In this study, we chose the commercial software, Aimsun, for the following reasons:

- Aimsun has embedded the EV model for “Car”.
 - <https://www.aimsun.com/technical-notes/mfc-and-battery-consumption-model/>
 - Note: SUMO has a module for EV and Charging Station.
https://sumo.dlr.de/docs/Models/Electric.html#stopping_at_a_charging_station
- Aimsun is one of the prevailing commercial microsimulation tools available in the market. It can import the networks from other commercial software tools like Vissim and Paramics.
- Commercial tools normally have better GUI support and more features available to support various needs of the stakeholders.
 - Aimsun has the Vehicle-to-Everything (V2X) module to further extend the evaluation platform developed in this study for Connected and Automated, Electric Vehicles (CAEVs).